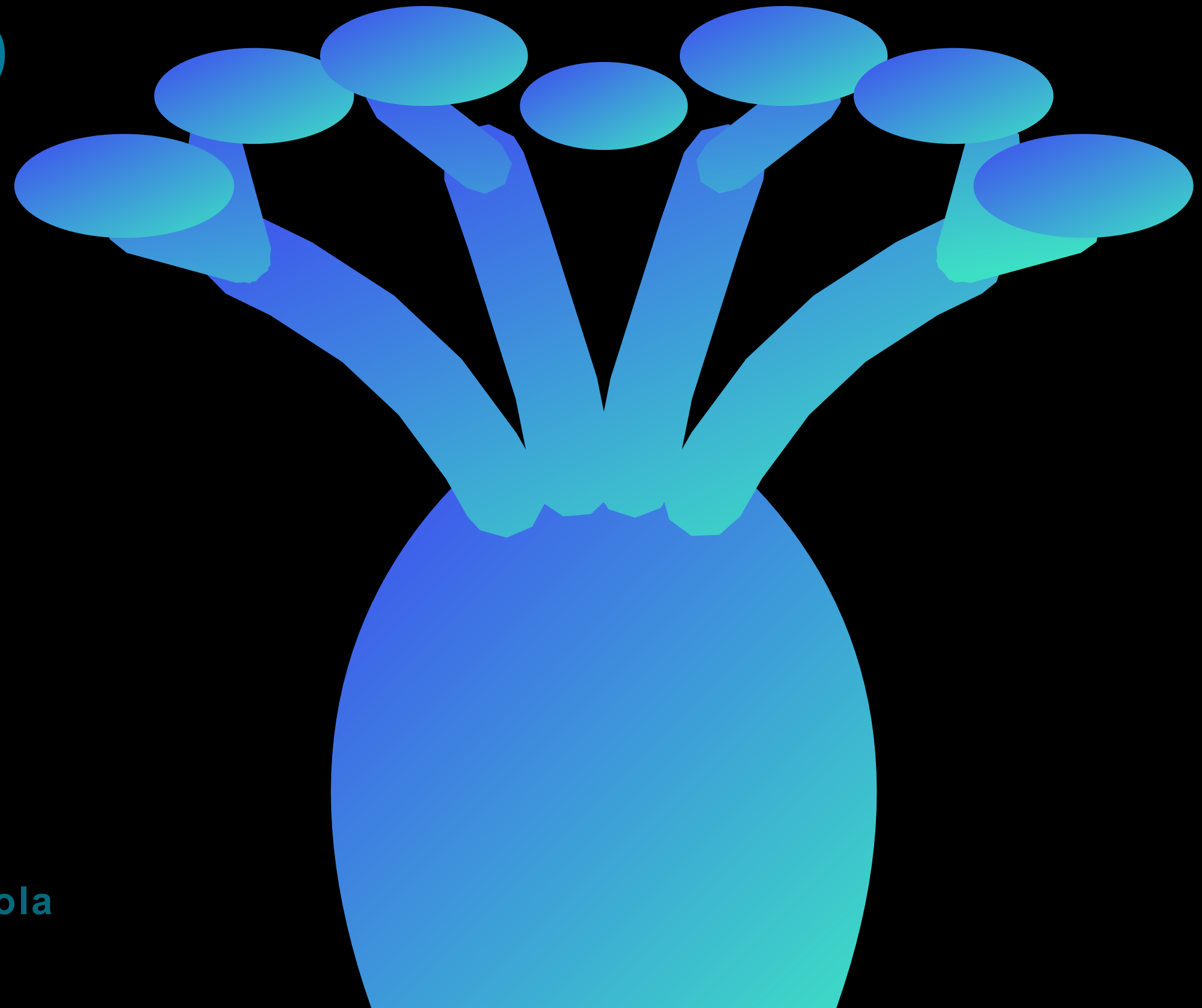


ECO4053W PROJECT · September 2026

BAOBAB

Rooted here. Reaching there.



**Group 3 - Karabo Tigedi, Kerry-Lynn Whyte, Nikola
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BUSINESS PROBLEM

\$200 remittances to Zimbabwe **cost 4x** the **UN**
target(3%)

Cheapest option

5.15%

(WorldRemit
Online)

Corridor average

12.49%

4x the SDG 3%
target

Priciest option

31.81%

Western Union,
Agent

**Sent formally,
SA->SADC (2024)**

R19 Billion

Mostly small, recurring
amounts

Stablecoins make settlement cheap, but the sender must buy it and the recipient must hold and sell it, as with VALR Pay.

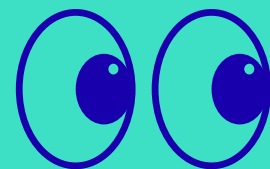
Our platform: pay in rand → settle on XRPL in RLUSD → cash out to local currency

PROPOSED SOLUTION

A transparent payment, settled on the XRP ledger

Every fee on the quote

A fixed rand fee plus a percentage and a published FX margin. The sender sees each part before paying.



A 15-minute locked quote

The price holds for 15 minutes. Cancelled or expired quotes stop counting toward the sender's limits.



A payment you can look up

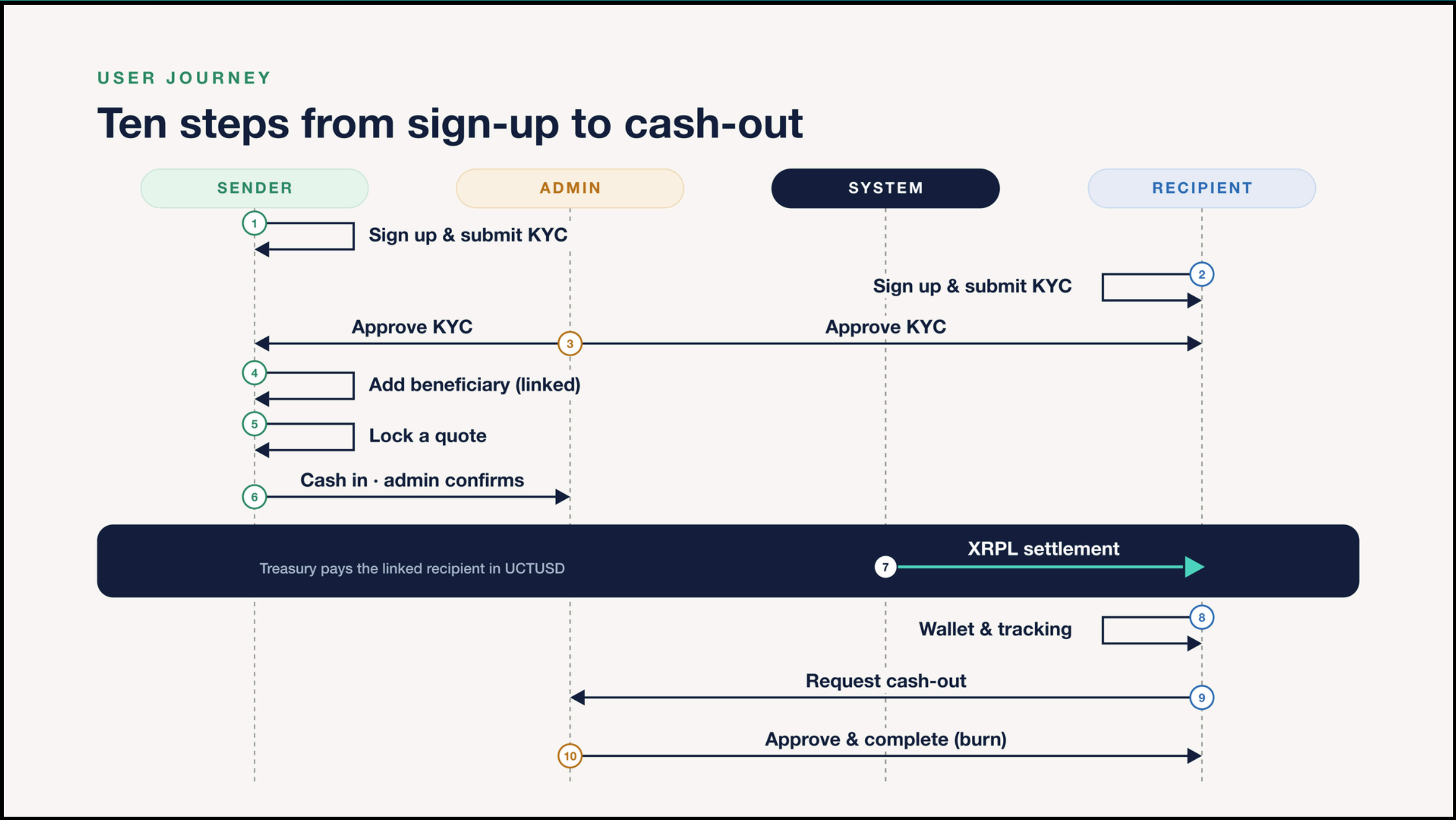
Each payout is a real XRPL Testnet Payment. The recipient gets its transaction hash.



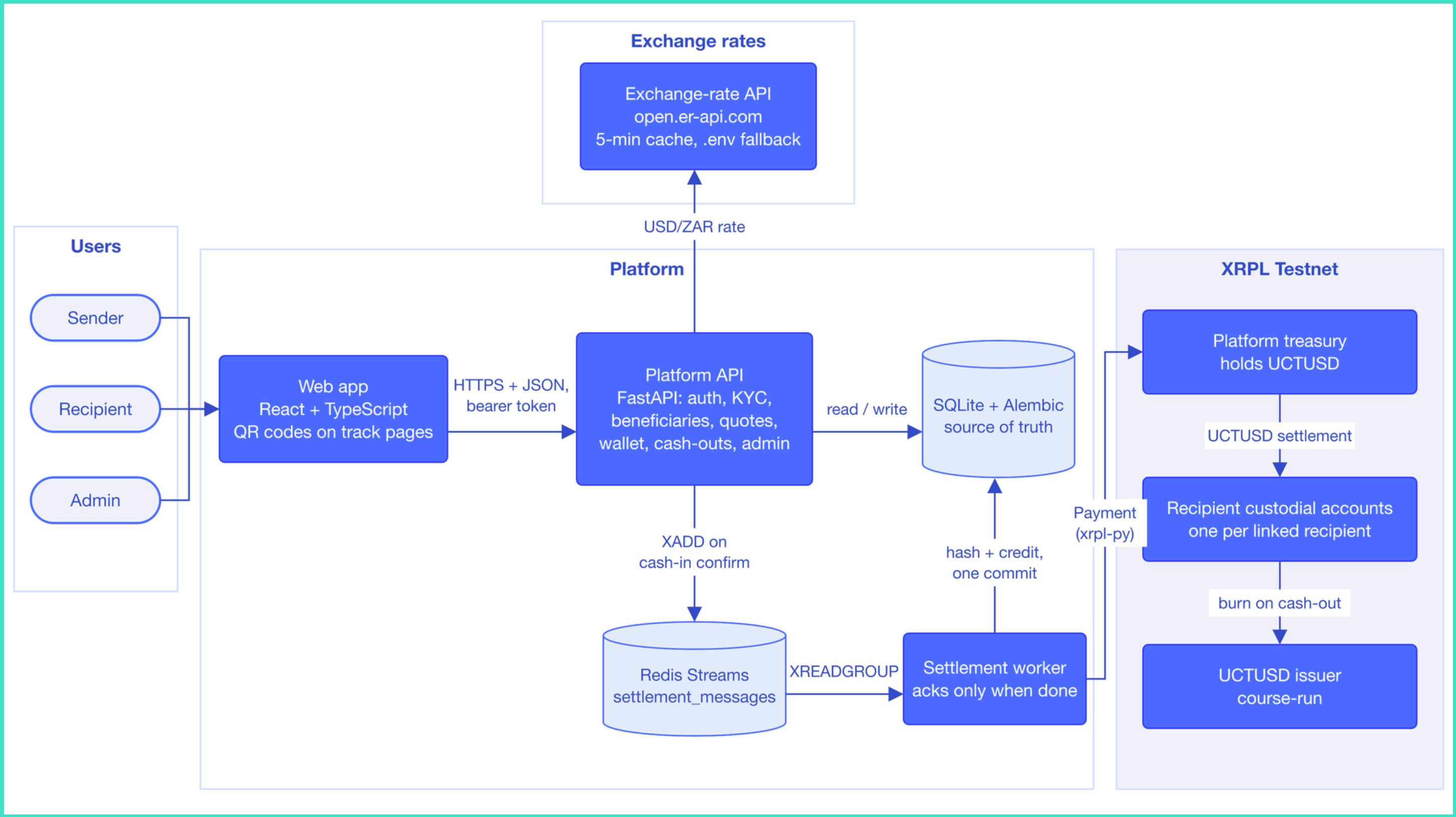
Settlement asset: UCTUSD, the course's issued token, on **XRPL Testnet**.

- Future work, can switch to RLUSD (a config change).
- **Cash-in and cash-out are simulated.**

USER JOURNEY



ARCHITECTURE



FEE MODEL

Every fee is visible before you commit



fee = R25 + 1% × send

rate = mid-market × 1.02

net = send - fee

UCTUSD = net ÷ rate

- Seeded defaults, editable by an admin:
 - R25 fixed, 1% of the send, 2% FX margin, 1.5% at cash-out.
- A sending amount that the fee would swallow is rejected.

Worked example:
R1,000 at 18.50
ZAR/USD

Fee	Amount
Transaction fee	R35.00
Effective rate	18.87
Net after fee	R965.00
Recipient receives	51.14 UCTUSD

SECURITY

1. Passwords and Sessions

- **Passwords** are hashed with bcrypt.
- **Session tokens** are stored only as **SHA-256 hashes**, and **logout revokes** them at once.

2. Two separate fernet keys

- One for **KYC ID numbers**, one for **XRPL secrets**.
- **Leaking one doesn't open the other**.
- A wallet key is decrypted only just before signing.

3. No secret key in any response

- **Wallet response models** have no **secret field**.
- **Keys live in environment variables**, never in git, logs or the API.

4. Passwords and Sessions

- **Customers act only on their own records**.
Sending and cash-out need approved KYC.
- **Admins are created by script**, never by sign-up.



REGULATORY CONSIDERATIONS

A stablecoin does not remove the rules

Area	A real operator needs	What we built
KYC and AML (FICA)	Due diligence, sanctions screening, STRs	Mock KYC, admin approval, encrypted ID
Monitoring	Velocity and structuring alerts	Daily and monthly rand caps; no monitoring engine
Crypto custody	FSCA crypto-asset authorisation	Custodial Testnet keys, encrypted
Exchange control	SARB authorised dealer; R2m annual allowance	Allowance not enforced; fiat legs simulated
Licensing	FICA, SARB and FSCA as applicable	Unlicensed; Testnet only, no real funds

TESTING

Stable under load; the ledger sets the pace

0 failures

in 2,218 requests
from 50 concurrent
users (37 req/s)

1.37s

slowest request;
the target was
under 2 s

15/15

real Testnet
payments settled,
15.1 s average

138

automated pytest
tests pass

Login is slow on purpose

- bcrypt puts login at about 850 ms median. Authenticated reads run at 20 - 33 milliseconds.

Settlement waits on consensus

- Payments took 12.4 - 16.2 seconds each, waiting for Testnet to validate.

LESSONS LEARNED

- **Acknowledge a job only when it's finished** - acking a settlement message on pickup meant a crash lost the job. Now the worker acks only after the job has completed or failed.
- **Retries must never pay twice** - a crash after paying could trigger a second Payment. The worker now checks for a stored hash and looks up the ledger by memo before it pays.
- **Blockchains add setup steps before any payment** - a wallet can't hold UCTUSD until its TrustLine exists, and a missing one fails with `tecPATH_DRY`. Every settlement now checks for the TrustLine before it pays.
- **The ledger sets the speed, not our code** - each Testnet payment took 12–16 seconds to validate. Benchmarking showed us not to waste time optimising our own code.
- **Protect secrets from your own team, too** - Our database holds the treasury's private key. It nearly got deleted, and later a copy was pushed to GitHub. Secrets stay out of git, and the database file gets treated like cash.

WHERE DOES IT GO NEXT



1. Known gaps

- **No row lock** between the limit check and the quote insert, so two parallel quotes could exceed a cap
- **Two overlapping worker runs could both miss the ledger lookup and pay twice**
- **SQLite in development**; no AML rules engine or sanctions screening
- **Cash-in and cash-out are simulated**

2. Next steps

- A **production database with locking on limit checks**
- A **lock so only one worker run pays at a time**
- **Exchange rates from the Kraken API; evaluate Celery or RQ**
- **Point settlement at official RLUSD once funded**

BAOBAB

Rooted here. Reaching there.

Thank you.

